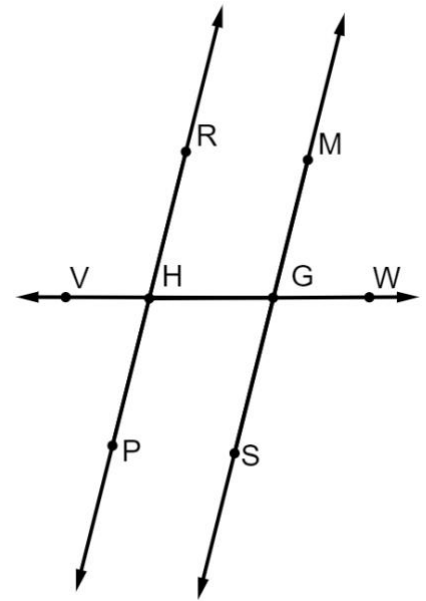


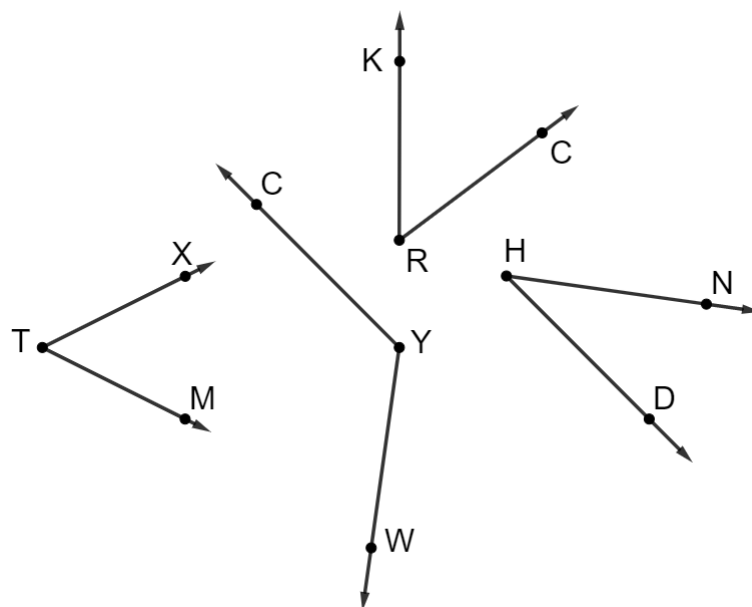
1. Lines RP , VW , and MS are shown. $\overleftrightarrow{VW}$ intersects $\overleftrightarrow{RP}$ at point H . $\overleftrightarrow{VW}$ intersects $\overleftrightarrow{MS}$ at point G .



First, determine if each condition is sufficient on its own to show that $\overleftrightarrow{RP} \parallel \overleftrightarrow{MS}$. Then, write a justification.

	Condition	Sufficient to show that $\overleftrightarrow{RP} \parallel \overleftrightarrow{MS}$?	Justification
a.	$\angle PHG \cong \angle MGH$	☐ Yes ☐ No	
b.	$\angle RHG$ and $\angle MGH$ are supplementary.	☐ Yes ☐ No	
c.	$\angle SGW \cong \angle PHV$	☐ Yes ☐ No	
d.	$m\angle WGM \cong m\angle GHR$	☐ Yes ☐ No	

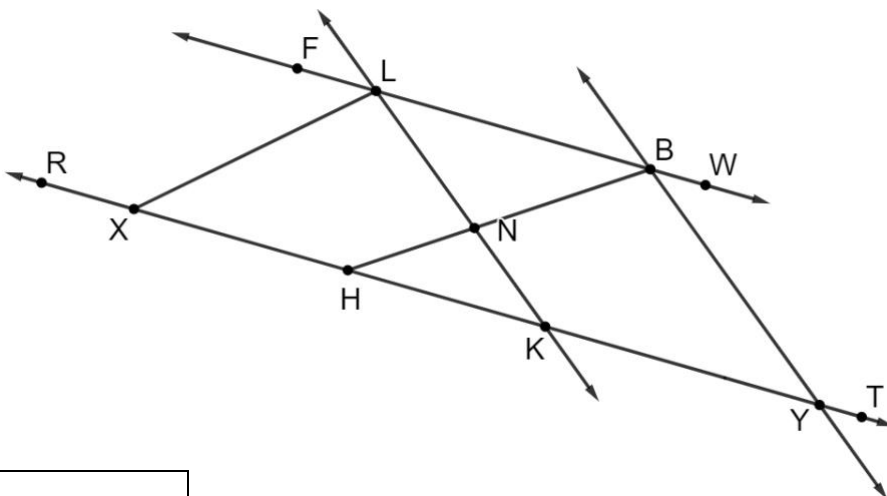
2. Four angles are shown where $\angle NHD$ is complementary to $\angle XTM$, $\angle KRC$ is supplementary to $\angle CYW$, and $\angle KRC$ is complementary to $\angle NHD$. $m\angle CYW = 11x - 27$, $m\angle XTM = 3x + 11$, and $m\angle NHD = 2x + 9$.



Select all of the statements that are true based on the given information.

- ☐ $m\angle XTM = m\angle KRC$
- ☐ $3x + 11 + 2x + 9 = 90^\circ$
- ☐ $11x - 27 + 2x + 9 = 180^\circ$
- ☐ $m\angle XTM + m\angle CYW = 180^\circ$
- ☐ $11x - 27 + 3x + 11 + 2x + 9 = 180^\circ$

3. A diagram is shown where $\overleftrightarrow{LK} \parallel \overleftrightarrow{BY}$, $\overleftrightarrow{FW} \parallel \overleftrightarrow{XT}$, $m\angle XLN = 100^\circ$, $m\angle NHK = 34^\circ$, and $m\angle YBH = 107^\circ$.



Complete the table.

a.

$$m\angle KYB = \underline{\hspace{2cm}}$$

b.

$$m\angle XLF = \underline{\hspace{2cm}}$$

4. Parallel lines $\overleftrightarrow{LN}$, $\overleftrightarrow{BH}$, and $\overleftrightarrow{RK}$, with a transversal $\overleftrightarrow{PW}$, are shown $m\angle BMV = 108^\circ$ and $m\angle KVS = 72^\circ$.

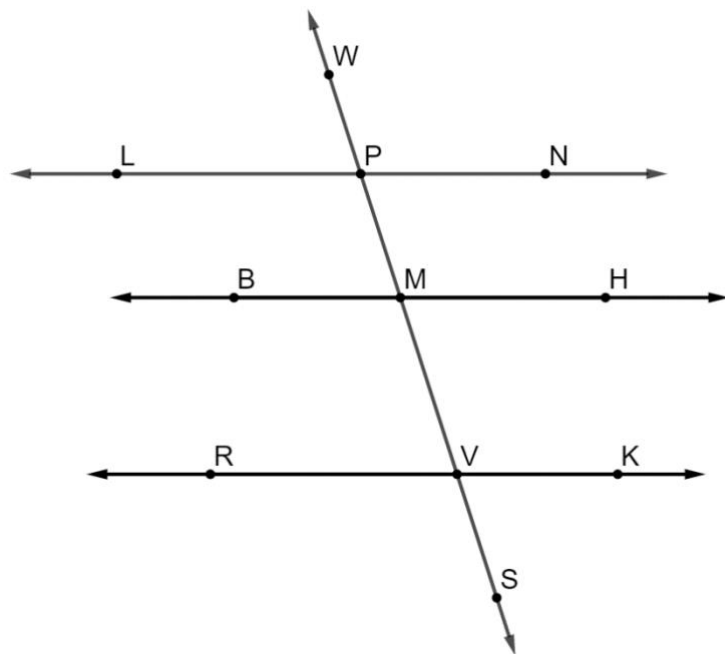
Complete the table.

a.

$$m\angle WPN = \underline{\hspace{2cm}}$$

b.

$$m\angle LPW = \underline{\hspace{2cm}}$$



5. Parallel lines $\overleftrightarrow{BW}$ and $\overleftrightarrow{DM}$ are shown where $m\angle XDM = 138^\circ$ and $m\angle HBW = 29^\circ$.

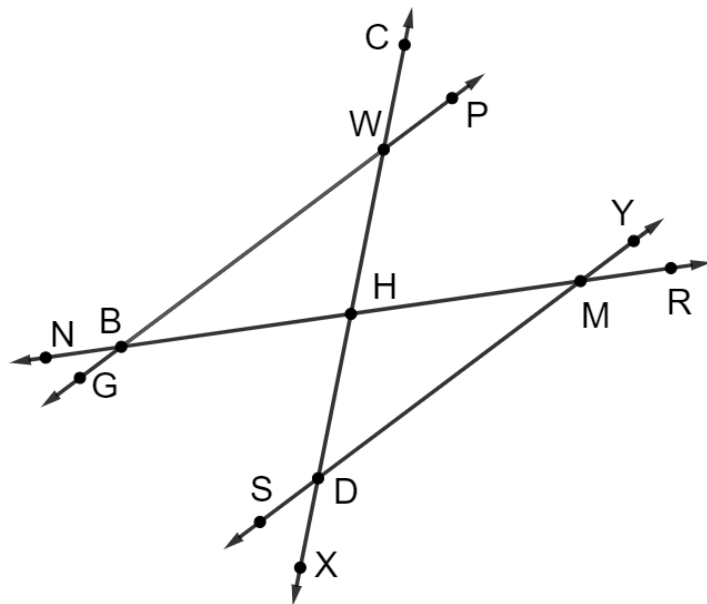
Complete the table.

a.

$$m\angle CWB = \underline{\hspace{2cm}}$$

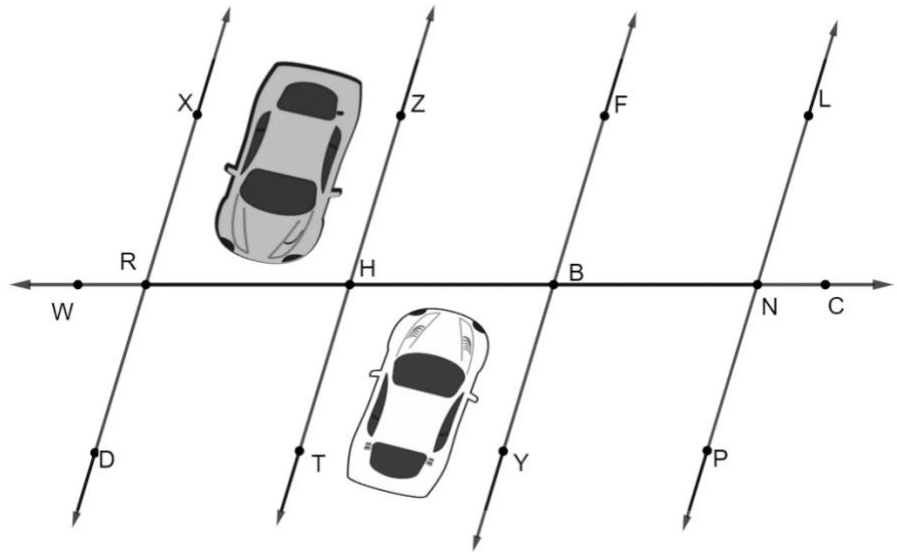
b.

$$m\angle HMY = \underline{\hspace{2cm}}$$



6. An engineer is designing a parking lot for a local grocery store. The parking spaces are marked with lines where $\overleftrightarrow{XD} \parallel \overleftrightarrow{TZ} \parallel \overleftrightarrow{YF} \parallel \overleftrightarrow{LP}$ where $\overleftrightarrow{WC}$ is a transversal, as shown, where $m\angle CNL = (6x - 17.6)^\circ$, $m\angle FBH = (12y - 22)^\circ$, and $m\angle THR = (3x + 29.2)^\circ$.

a. Determine the value of x .



$x =$	
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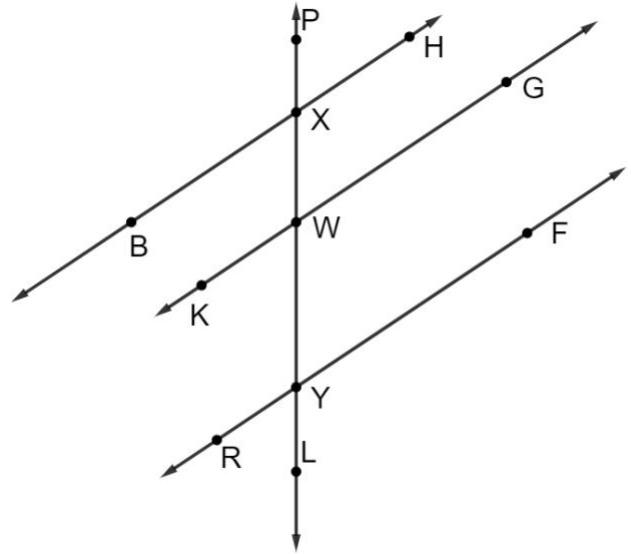
b. Determine the value of y .

$y =$	
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7. A diagram is shown where $\overleftrightarrow{PL}$ intersects lines BH , KG , and RF .

Given: $\overleftrightarrow{BH} \parallel \overleftrightarrow{RF}$, $\angle HXW \cong \angle YWG$

Prove: $\overleftrightarrow{RF} \parallel \overleftrightarrow{KG}$



Complete the proof.

Statement	Reason
1. $\overleftrightarrow{BH} \parallel \overleftrightarrow{RF}$	1. Given
2.	2.
3. $\angle HXW \cong \angle YWG$	3. Given
4.	4.
5. $\overleftrightarrow{RF} \parallel \overleftrightarrow{KG}$	5.